

Ejercicio 1.

Modelo:

$$f(x, \theta) = \omega x + \beta \quad \theta = \{\omega, \beta\}$$

1.1 Forward pass:

$$z_1 = \omega x$$

$$z_2 = z_1 + \beta$$

Salida:

$$f = z_2$$

1.2. Loss:

$$\ell = (f - y)^2$$

1.3. Forward computational graph

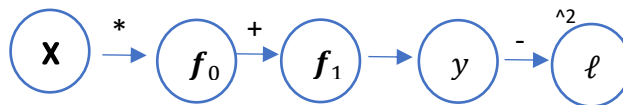
Forward pass:

$$f_0 = \omega x$$

$$f_1 = f_0 + \beta$$

$$y = f_1$$

$$\ell = (f - y)^2$$



1.4. Backward Pass:

$$\frac{\partial \ell}{\partial f} = 2(f - y)$$

Ahora

$$z_1 = \omega x$$

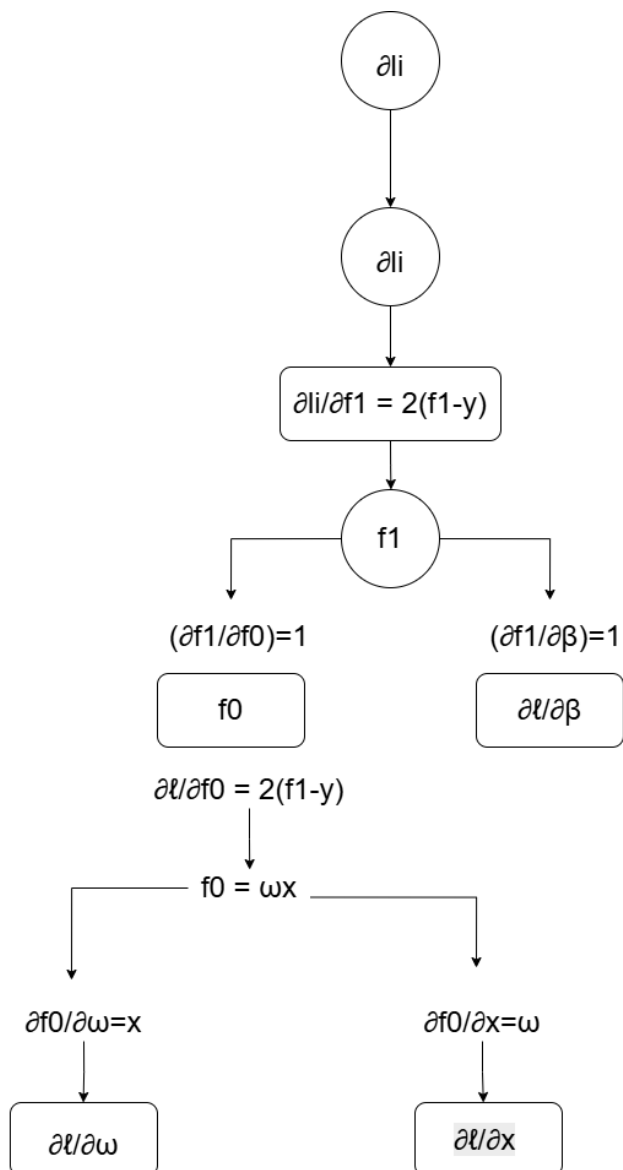
por lo tanto

$$\frac{\partial z_1}{\partial \omega} = x \quad y \quad \frac{\partial z_1}{\partial x} = \omega$$

Finalmente

$$\frac{\partial \ell}{\partial \omega} = 2(f - y) x \quad y \quad \frac{\partial \ell}{\partial \beta} = 2(f - y)$$

1.5.- Backward computational graph



1.6.- Compute the gradients with respect to all learnable parameters in θ .

$$\theta = \{\omega, \beta\}$$

$$\frac{\partial \ell}{\partial \omega} = 2 (f - y) \times \quad y \quad \frac{\partial \ell}{\partial \beta} = 2 (f - y)$$

Sustituyendo

$$f = \omega x + \beta$$

Es igual a:

$$\frac{\partial \ell}{\partial \omega} = 2 (\omega x + \beta - y) \times \quad y \quad \frac{\partial \ell}{\partial \beta} = 2 (\omega x + \beta - y)$$

Ejercicio 2. One Nonlinear Activation

Model: $f(x, \theta) = \tanh(\omega x + \beta)$ $\theta = (\omega, \beta)$

2.1 Forward pass:

$$f_0 = \omega x$$

$$f_1 = f_0 + \beta$$

Salida

$$f = \tanh(z_2)$$

2.2 Loss

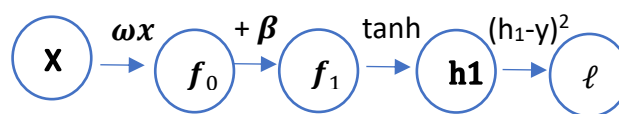
$$\ell = (f - y)^2$$

$$f = \tanh(\omega x + \beta)$$

$$\ell = (\tanh(\omega x + \beta) - y)^2$$

2.3 Forward computational graph

$$\begin{aligned} f_0 &= \omega x \\ f_1 &= f_0 + \beta \\ h1 &= \tanh(f_1) \\ \ell &= (h1 - y)^2 \end{aligned}$$



2.4 Backward pass

$$\ell = (h1 - y)^2$$

derivando

$$\frac{\partial \ell}{\partial f} = 2(h_1 - y)$$

$$h_1 = \tanh(f_1)$$

$$\frac{d}{df_1} \tanh(f_1) = 1 - \tanh^2(f_1)$$

$$h_1 = \tanh(f_1)$$

entonces

$$\frac{\partial h_1}{\partial f_1} = 1 - h_1^2$$

Regla de la cadena

$$\frac{\partial \ell}{\partial f_1} = 2(h_1 - y)(1 - h_1^2)$$

Suma

$$f_1 = f_0 + \beta$$

$$\frac{\partial f_1}{\partial f_0} = 1$$

$$\frac{\partial f_1}{\partial \beta} = 1$$

$$\frac{\partial \ell}{\partial \beta} = \frac{\partial \ell}{\partial f_1} \cdot \frac{\partial f_1}{\partial \beta} = 2(h_1 - y)(1 - h_1^2)$$

$$\frac{\partial \ell}{\partial f_0} = \frac{\partial \ell}{\partial f_1} \cdot \frac{\partial f_1}{\partial f_0} = 2(h_1 - y)(1 - h_1^2)$$

Considerando que

$$f_0 = \omega x$$

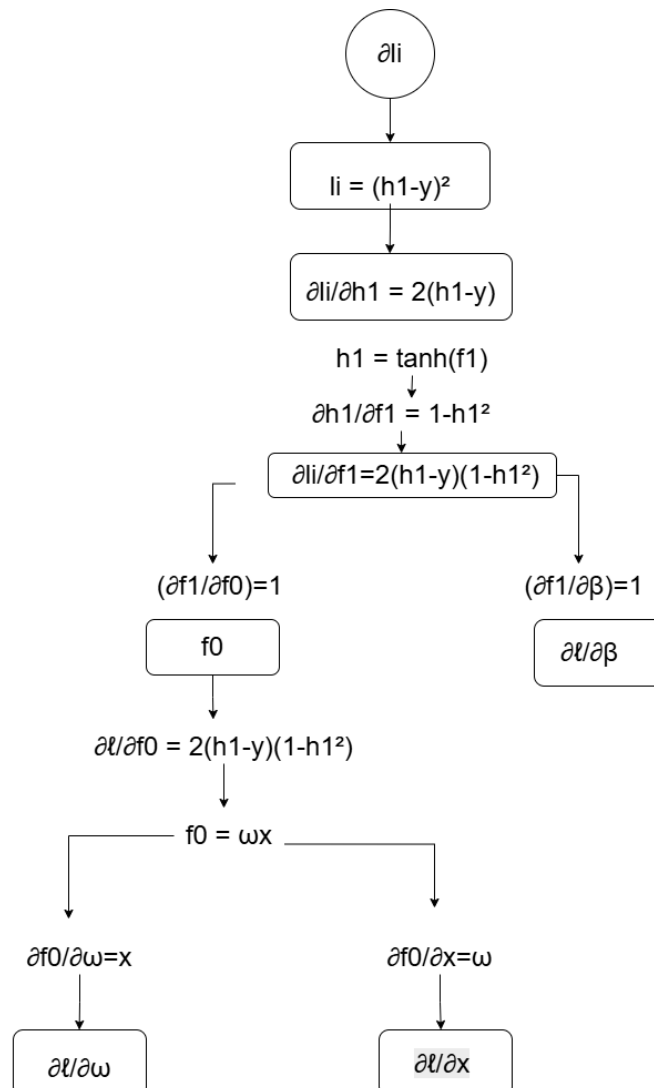
$$\frac{\partial f_0}{\partial \omega} = x$$

$$\frac{\partial f_0}{\partial x} = \omega$$

Regla de la cadena

$$\frac{\partial \ell}{\partial \omega} = \frac{\partial \ell}{\partial f_0} \cdot \frac{\partial f_0}{\partial \omega} = 2(h_1 - y)(1 - h_1^2)x$$

2.5 Backward computational graph



2.6.- Compute the gradients with respect to all learnable parameters.

$$\theta = \{\omega, \beta\}$$

$$\frac{\partial \ell}{\partial \beta} = 2(h_1 - y)(1 - h_1^2)$$

$$\frac{\partial \ell}{\partial \omega} = 2(h_1 - y)(1 - h_1^2)\mathbf{x}$$

Exercise 3: One Hidden Neuron with Sine

Model:

$$f(x, \theta) = \omega_1 \sin(\omega_0 x + \beta_0) + \beta_1$$

$$\theta = \omega_0, \beta_0, \omega_1, \beta_1$$

3.1 Forward Pass

$$f_0 = \omega_0 x$$

Suma

$$f_1 = f_0 + \beta_0$$

Seno

$$h_1 = \sin(f_1)$$

Multiplicacion

$$f_2 = \omega_1 h_1$$

Entonces

$$f_3 = f_2 + \beta_1$$

Por lo tanto

3.2 Loss:

$$\ell = (f_3 - y)^2$$

3.3. Forward Computational Graph

$$f_0 = \omega_0 x$$

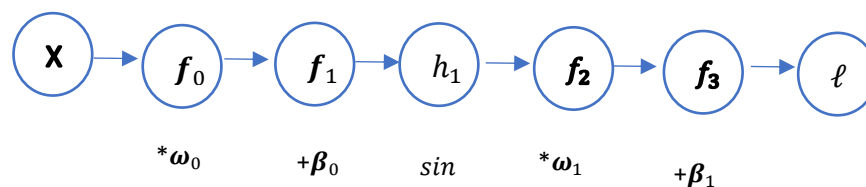
$$f_1 = f_0 + \beta_0$$

$$h_1 = \sin(f_1)$$

$$f_2 = \omega_1 h_1$$

$$f_3 = f_2 + \beta_1$$

$$\ell = (f_3 - y)^2$$



3.4 Backward Pass

$$\ell = (f_3 - y)^2$$

$$\frac{\partial \ell}{\partial f_3} = 2(f_3 - y)$$

Tomando en cuenta que

$$f_3 = f_2 + \beta_1$$

$$\frac{\partial f_3}{\partial f_2} = 1$$

$$\frac{\partial f_3}{\partial \beta_1} = 1$$

Regla de la cadena

$$\frac{\partial \ell}{\partial f_2} = 2(f_3 - y)$$

$$\frac{\partial \ell}{\partial \beta_1} = 2(f_3 - y)$$

Multiplicación

$$f_2 = \omega_1 h_1$$

$$\frac{\partial f_2}{\partial \omega_1} = h_1$$

$$\frac{\partial f_2}{\partial h_1} = \omega_1$$

Regla de la cadena

$$\frac{\partial \ell}{\partial \omega_1} = 2(f_3 - y)h_1$$

$$\frac{\partial \ell}{\partial h_1} = 2(f_3 - y)\omega_1$$

Fucion seno

$$h_1 = \sin(f_1)$$

$$\frac{\partial h_1}{\partial f_1} = \cos(f_1)$$

Regla de la cadena

$$\frac{\partial \ell}{\partial f_1} = 2(f_3 - y)\omega_1 \cos(f_1)$$

Suma:

$$f_1 = f_0 + \beta_0$$

$$\frac{\partial f_1}{\partial f_0} = 1$$

$$\frac{\partial f_1}{\partial \beta_0} = 1$$

Regla de la cadena

$$\frac{\partial \ell}{\partial \beta_0} = 2(f_3 - y)\omega_1 \cos(f_1)$$

$$\frac{\partial \ell}{\partial f_0} = 2(f_3 - y)\omega_1 \cos(f_1)$$

Multiplicacion

$$f_0 = \omega_0 x$$

Entonces

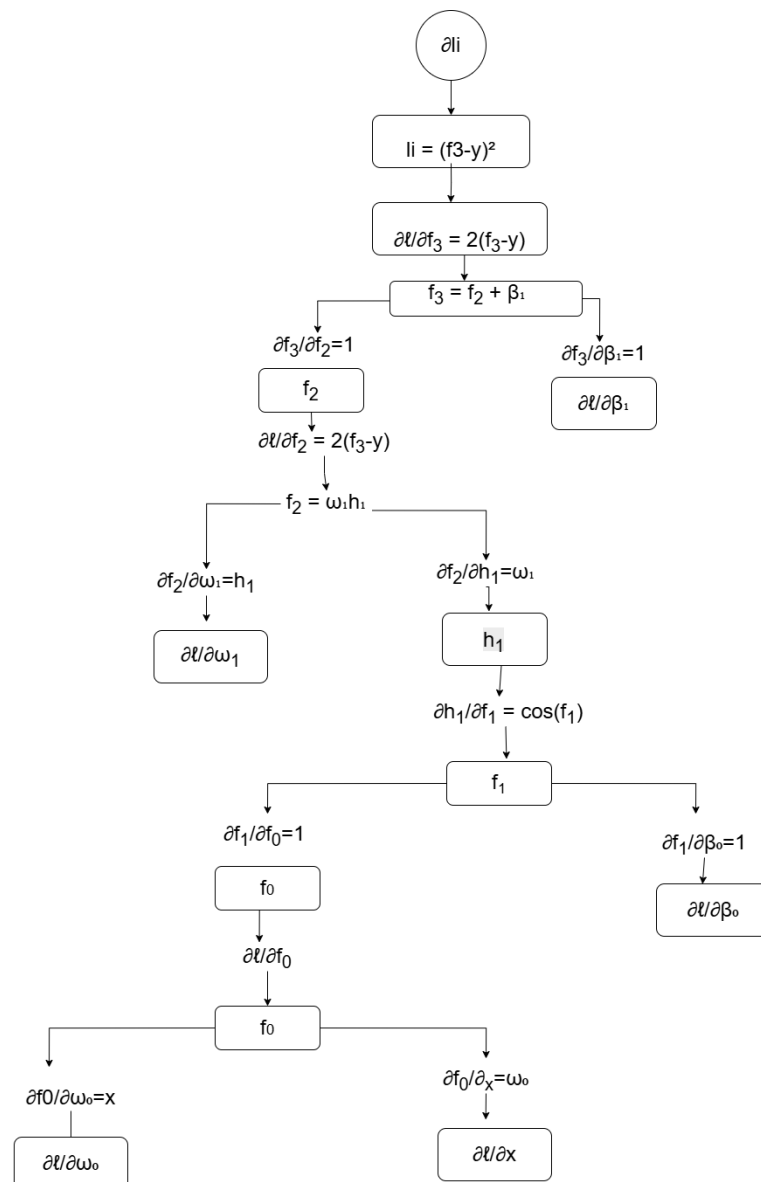
$$\frac{\partial f_0}{\partial \omega_0} = x$$

$$\frac{\partial f_0}{\partial x} = \omega_0$$

Regla de la cadena

$$\frac{\partial \ell}{\partial \omega_0} = 2(f_3 - y)\omega_1 \cos(f_1)x$$

3.5 Backward Computational Graph



3.6 Compute the gradients with respect to all learnable parameters in θ

$$\theta = \omega_0, \beta_0, \omega_1, \beta_1$$

$$\frac{\partial \ell}{\partial \omega_0} = 2(f_3 - y)\omega_1 \cos(f_1)x$$

$$\frac{\partial \ell}{\partial \beta_0} = 2(f_3 - y)\omega_1 \cos(f_1)$$

$$\frac{\partial \ell}{\partial \omega_1} = 2(f_3 - y)h_1$$

$$\frac{\partial \ell}{\partial \beta_1} = 2(f_3 - y)$$

Exercise 4: Scalar Deep MLP

Model:

$$f(x, \theta) = \omega_2 \tanh(\omega_1 \sin(\omega_0 x + \beta_0) + \beta_1) + \beta_2$$

$$\theta = \omega_0, \beta_0, \omega_1, \beta_1, \omega_2, \beta_2$$

4.1 Forward Pass

$$f_0 = \omega_0 x$$

$$f_1 = f_0 + \beta_0$$

Seno

$$h_1 = \sin(f_1)$$

Multiplicación

$$f_2 = \omega_1 h_1$$

Suma

$$f_3 = f_2 + \beta_1$$

Tangente

$$h_2 = \tanh(f_3)$$

Multiplicación

$$f_4 = \omega_2 h_2$$

Suma

$$f_5 = f_4 + \beta_2$$

4.2 Loss

$$\ell = (f_5 - y)^2$$

4.3. Forward Computational Graph

$$f_0 = \omega_0 x$$

$$f_1 = f_0 + \beta_0$$

$$h_1 = \sin(f_1)$$

$$f_2 = \omega_2 h_1$$

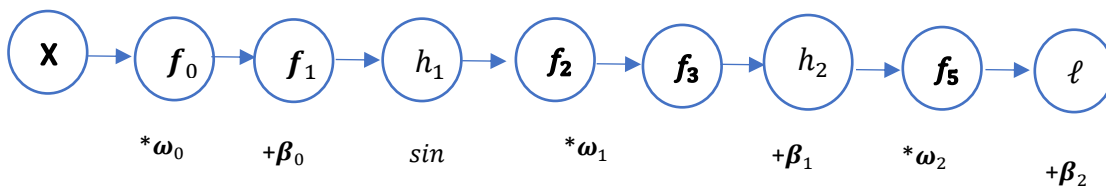
$$f_3 = f_2 + \beta_1$$

$$h_2 = \tanh(f_3)$$

$$f_4 = \omega_2 h_2$$

$$f_5 = f_4 + \beta_2$$

$$\ell = (f_5 - y)^2$$



4.4 Backward Pass

$$\ell = (f_5 - y)^2$$

$$\frac{\partial \ell}{\partial f_5} = 2(f_5 - y)$$

Derivada de la suma final

$$f_5 = f_4 + \beta_2$$

$$\frac{\partial f_5}{\partial f_4} = 1$$

$$\frac{\partial f_5}{\partial \beta_2} = 1$$

Regla de la cadena

$$\frac{\partial \ell}{\partial f_4} = 2(f_5 - y)$$

$$\frac{\partial \ell}{\partial \beta_2} = 2(f_5 - y)$$

Multiplicación

Tomando en cuenta que

$$f_4 = \omega_2 h_2$$

Las derivadas son

$$\frac{\partial f_4}{\partial \omega_2} = h_2$$

$$\frac{\partial f_4}{\partial h_2} = \omega_2$$

Regla de la cadena

$$\frac{\partial \ell}{\partial \omega_2} = 2(f_5 - y)h_2$$

$$\frac{\partial \ell}{\partial h_2} = 2(f_5 - y)\omega_2$$

Tanh

$$h_2 = \tanh(f_3)$$

Derivada

$$\frac{\partial h_2}{\partial f_3} = 1 - h_2^2$$

Regla de la cadena

$$\frac{\partial \ell}{\partial f_3} = 2(f_5 - y)\omega_2(1 - h_2^2)$$

Suma

$$f_3 = f_2 + \beta_1$$

$$\frac{\partial f_3}{\partial f_2} = 1$$

$$\frac{\partial f_3}{\partial \beta_1} = 1$$

Regla de la cadena

$$\frac{\partial \ell}{\partial \beta_1} = 2(f_5 - y)\omega_2(1 - h_2^2)$$

$$\frac{\partial \ell}{\partial f_2} = 2(f_5 - y)\omega_2(1 - h_2^2)$$

Multiplicación

Tomando en cuenta que

$$f_2 = \omega_1 h_1$$

Las derivadas son

$$\frac{\partial f_2}{\partial \omega_1} = h_1$$

$$\frac{\partial f_2}{\partial h_1} = \omega_1$$

Regla de la cadena

$$\frac{\partial \ell}{\partial \omega_1} = 2(f_5 - y)\omega_2(1 - h_2^2)h_1$$

$$\frac{\partial \ell}{\partial h_1} = 2(f_5 - y)\omega_2(1 - h_2^2)\omega_1$$

Funcion seno

$$h_1 = \sin(f_1)$$

$$\frac{\partial h_1}{\partial f_1} = \cos(f_1)$$

Regla de la cadena

$$\frac{\partial \ell}{\partial f_1} = 2(f_5 - y)\omega_2(1 - h_2^2)\omega_1 \cos(f_1)$$

Suma

$$f_1 = f_0 + \beta_0$$

$$\frac{\partial f_1}{\partial f_0} = 1$$

$$\frac{\partial f_1}{\partial \beta_0} = 1$$

Regla de la cadena

$$\frac{\partial \ell}{\partial \beta_0} = 2(f_5 - y)\omega_2(1 - h_2^2)\omega_1 \cos(f_1)$$

$$\frac{\partial \ell}{\partial f_0} = 2(f_5 - y)\omega_2(1 - h_2^2)\omega_1 \cos(f_1)$$

Multiplicación

$$f_0 = \omega_0 x$$

Entonces

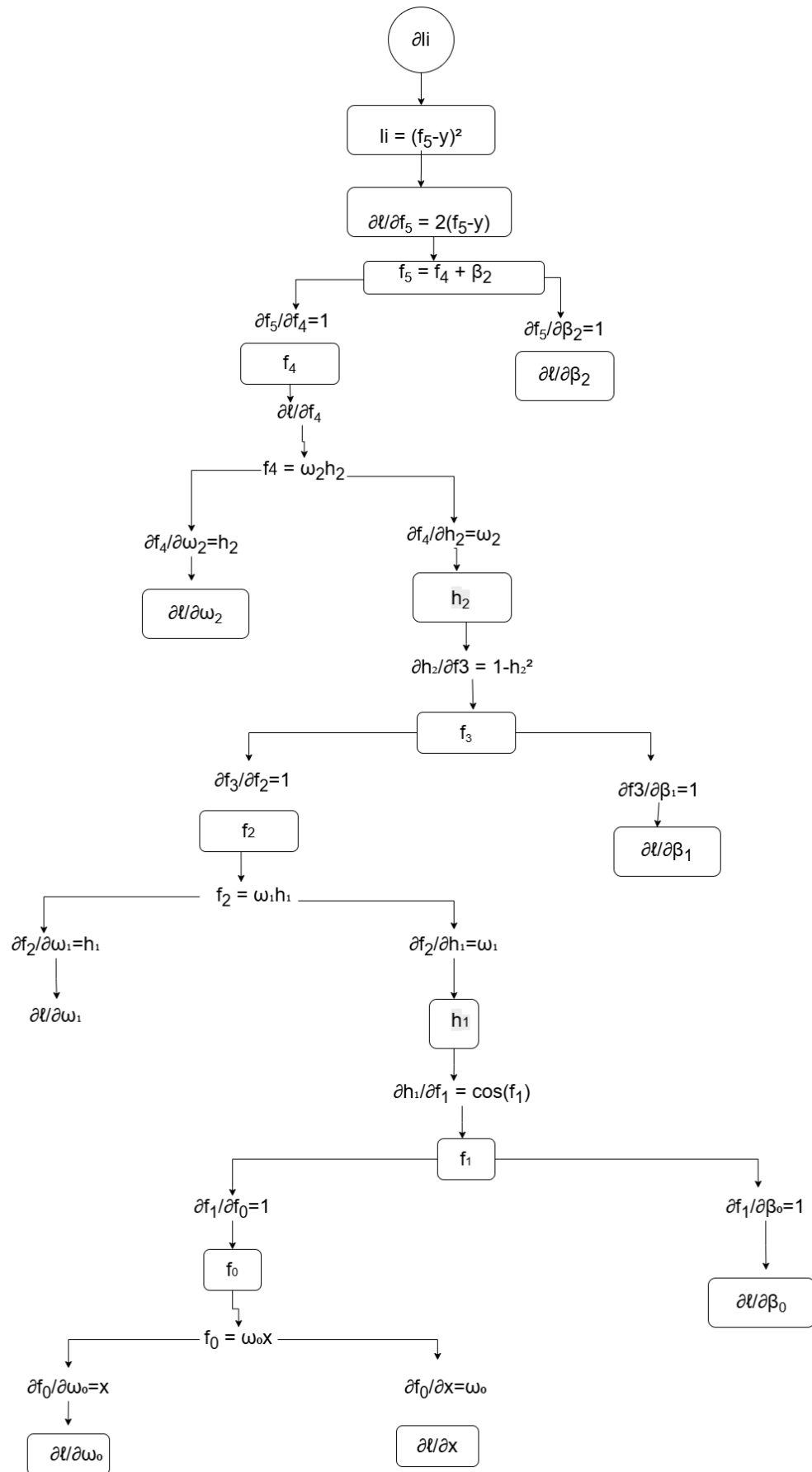
$$\frac{\partial f_0}{\partial \omega_0} = x$$

$$\frac{\partial f_0}{\partial x} = \omega_0$$

Regla de la cadena

$$\frac{\partial \ell}{\partial \omega_0} = 2(f_5 - y)\omega_2(1 - h_2^2)\omega_1 \cos(f_1)x$$

4.5 Backward Computational Graph



4.6 Compute the gradients with respect to all learnable parameters in θ

$$\theta = \omega_0, \beta_0, \omega_1, \beta_1, \omega_2, \beta_2$$

$$\frac{\partial \ell}{\partial \omega_0} = 2(f_5 - y)\omega_2(1 - h_2^2)\omega_1 \cos(f_1)x$$

$$\frac{\partial \ell}{\partial \beta_0} = 2(f_5 - y)\omega_2(1 - h_2^2)\omega_1 \cos(f_1)$$

$$\frac{\partial \ell}{\partial \omega_1} = 2(f_5 - y)\omega_2(1 - h_2^2)h_1$$

$$\frac{\partial \ell}{\partial \beta_1} = 2(f_5 - y)\omega_2(1 - h_2^2)$$

$$\frac{\partial \ell}{\partial \omega_2} = 2(f_5 - y)h_2$$

$$\frac{\partial \ell}{\partial \beta_2} = 2(f_5 - y)$$

Exercise 5: Two Hidden Neurons, One Input

Model: $f(x, \theta) = \omega_4 \tanh(\omega_0 x + \beta_0) + \omega_5 \sin(\omega_1 x + \beta_1) + \beta_2$

$$\theta = \omega_0, \beta_0, \omega_1, \beta_1, \omega_4, \omega_5, \beta_2$$

5.1 Forward Pass

Rama1 (tanh)

$$f_0 = \omega_0 x$$

$$f_1 = f_0 + \beta_0$$

$$h_1 = \tanh(f_1)$$

Multiplicación

$$f_2 = \omega_4 h_1$$

Rama2 (sin)

$$f_3 = \omega_1 x$$

$$f_4 = f_3 + \beta_1$$

$$h_2 = \sin(f_4)$$

Multiplicación

$$f_5 = \omega_5 h_2$$

Union de ramas

$$f_6 = f_2 + f_5$$

Suma total

$$f_7 = f_6 + \beta_2$$

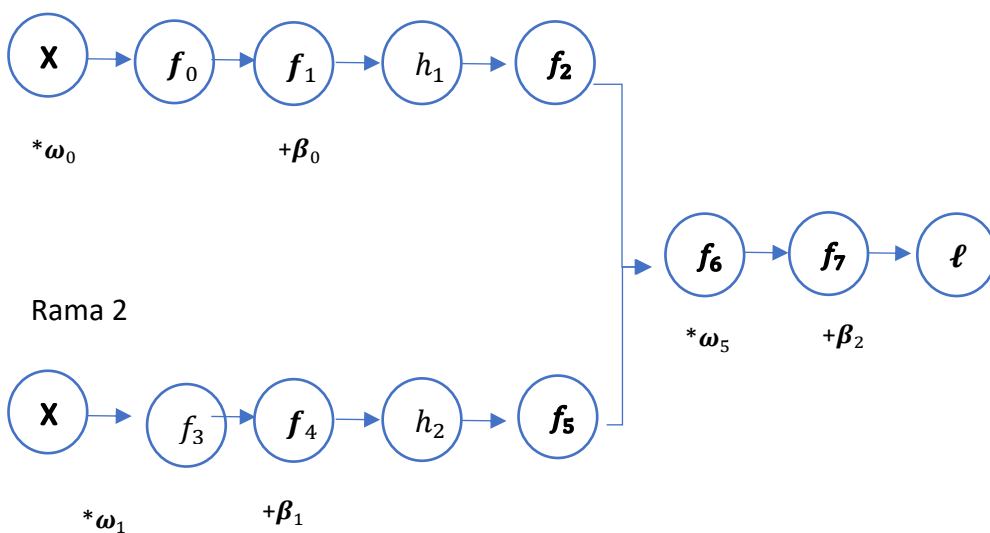
5.2 Loss

$$\ell = (f_7 - y)^2$$

5.3. Forward Computational Graph

$$\begin{aligned}f_0 &= \omega_0 x \\f_1 &= f_0 + \beta_0 \\h_1 &= \tanh(f_1) \\f_2 &= \omega_4 h_1 \\f_3 &= \omega_1 x \\f_4 &= f_3 + \beta_1 \\h_2 &= \sin(f_4) \\f_5 &= \omega_5 h_2 \\f_6 &= f_2 + f_5 \\f_7 &= f_6 + \beta_2 \\\ell &= (f_7 - y)^2\end{aligned}$$

Rama 1



5.4 Backward Pass

$$\ell = (f_7 - y)^2$$

Derivando

$$\frac{\partial \ell}{\partial f_7} = 2(f_7 - y)$$

Suma

$$f_7 = f_6 + \beta_2$$

$$\frac{\partial f_3}{\partial f_2} = 1$$

$$\frac{\partial f_3}{\partial \beta_1} = 1$$

Regla de la cadena

$$\frac{\partial \ell}{\partial f_6} = 2(f_7 - y)$$

$$\frac{\partial \ell}{\partial \beta_2} = 2(f_7 - y)$$

Separación de ramas

$$f_6 = f_2 + f_5$$

$$\frac{\partial f_6}{\partial f_2} = 1$$

$$\frac{\partial f_6}{\partial f_5} = 1$$

Por lo tanto

$$\frac{\partial \ell}{\partial f_2} = 2(f_7 - y)$$

$$\frac{\partial \ell}{\partial f_5} = 2(f_7 - y)$$

$$f_2 = \omega_4 h_1$$

$$\frac{\partial f_2}{\partial \omega_4} = h_1$$

$$\frac{\partial f_2}{\partial h_1} = \omega_4$$

Regla de la cadena

$$\frac{\partial \ell}{\partial \omega_4} = 2(f_7 - y)h_1$$

$$\frac{\partial \ell}{\partial h_1} = 2(f_7 - y)\omega_4$$

Tanh

$$h_1 = \tanh(f_1)$$

Derivada

$$\frac{\partial h_1}{\partial f_1} = 1 - h_1^2$$

Regla de la cadena

$$\frac{\partial \ell}{\partial f_1} = 2(f_7 - y)\omega_4(1 - h_1^2)$$

Suma

$$f_1 = f_0 + \beta_0$$

$$\frac{\partial \ell}{\partial \beta_0} = 2(f_7 - y)\omega_4(1 - h_1^2)$$

$$\frac{\partial \ell}{\partial f_0} = 2(f_7 - y)\omega_4(1 - h_1^2)$$

Regla de la cadena

$$\frac{\partial \ell}{\partial \beta_1} = 2(f_7 - y)\omega_4(1 - h_1^2)$$

$$\frac{\partial \ell}{\partial f_2} = 2(f_7 - y)\omega_4(1 - h_1^2)$$

Multiplificacion

$$f_0 = \omega_0 x$$

$$\frac{\partial \ell}{\partial \omega_0} = 2(f_7 - y)\omega_4(1 - h_2^2)x$$

Rama 2 (sin)

$$f_0 = \omega_5 x h_2$$

$$\frac{\partial f_5}{\partial \omega_5} = h_2$$

$$\frac{\partial f_5}{\partial h_2} = \omega_5$$

Regla de la cadena

$$\frac{\partial \ell}{\partial \omega_5} = 2(f_7 - y)h_2$$

$$\frac{\partial \ell}{\partial h_2} = 2(f_7 - y)\omega_5$$

Funcion seno

$$h_2 = \sin(f_4)$$

$$\frac{\partial h_2}{\partial f_4} = \cos(f_4)$$

Regla de la cadena

$$\frac{\partial \ell}{\partial f_1} = 2(f_7 - y)\omega_5 \cos(f_4)$$

Suma

$$f_4 = f_3 + \beta_1$$

$$\frac{\partial \ell}{\partial \beta_1} = 2(f_7 - y)\omega_5 \cos(f_4)$$

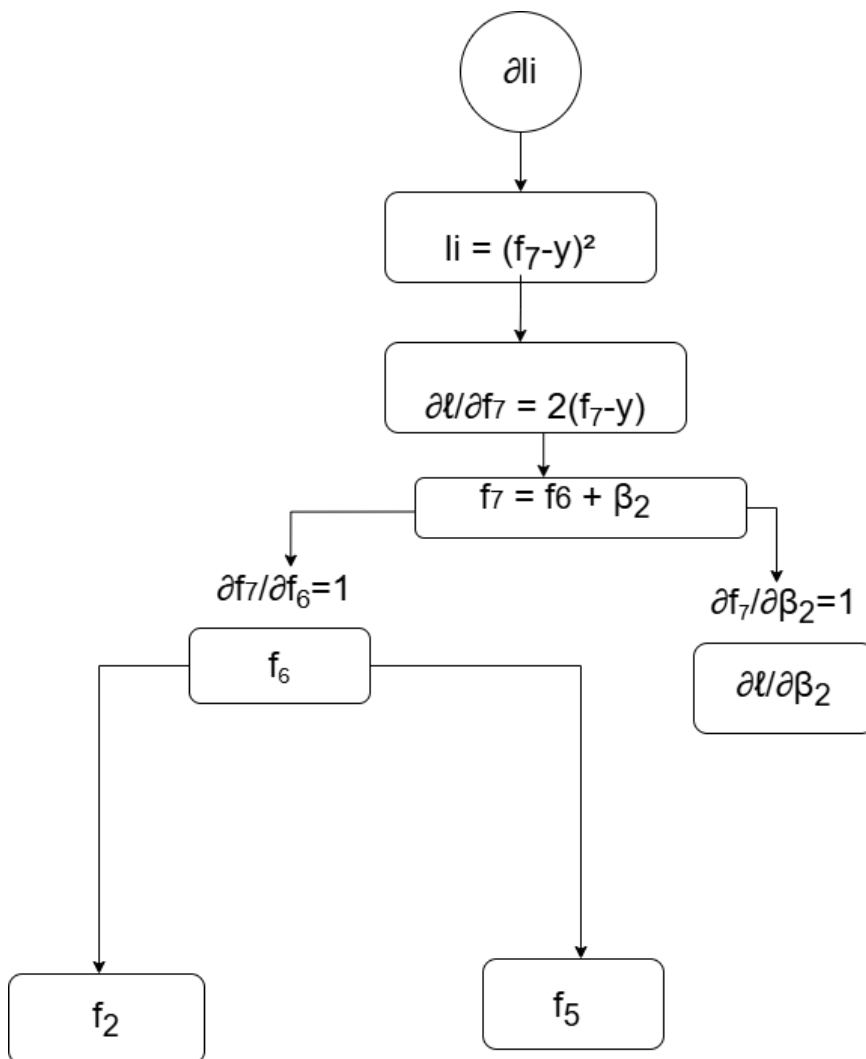
$$\frac{\partial \ell}{\partial f_3} = 2(f_7 - y)\omega_5 \cos(f_4)$$

Multiplicacion

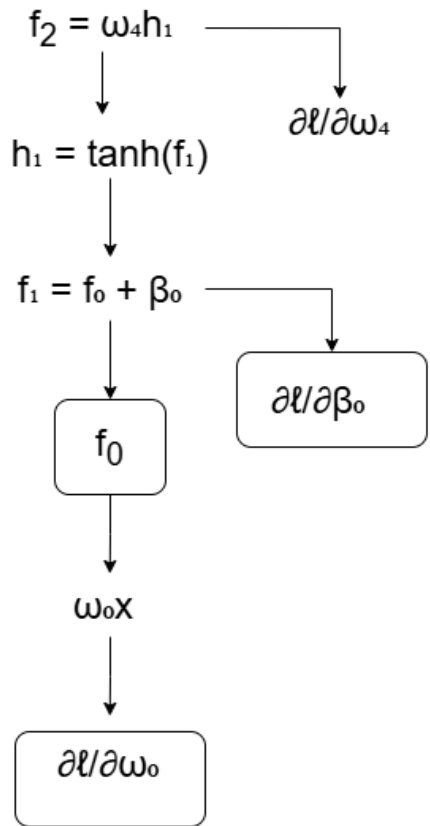
$$f_3 = \omega_1 x$$

$$\frac{\partial \ell}{\partial \omega_1} = 2(f_7 - y)\omega_5 \cos(f_4)x$$

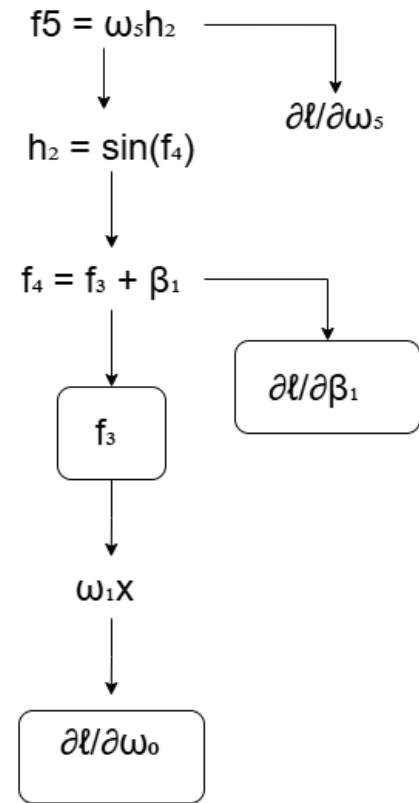
5.5 Backward Computational Graph



Rama 1



Rama 2



5.6 Compute the gradients with respect to all learnable parameters in θ

$$\theta = \omega_0, \beta_0, \omega_1, \beta_1, \omega_4, \omega_5, \beta_2$$

Gadrientes finales

$$\frac{\partial \ell}{\partial \omega_0} = 2(f_7 - y)\omega_4(1 - h_2^2)x$$

$$\frac{\partial \ell}{\partial \beta_0} = 2(f_7 - y)\omega_4(1 - h_1^2)$$

$$\frac{\partial \ell}{\partial \omega_1} = 2(f_7 - y)\omega_5 \cos(f_4)x$$

$$\frac{\partial \ell}{\partial \beta_1} = 2(f_7 - y)\omega_5 \cos(f_4)$$

$$\frac{\partial \ell}{\partial \omega_4} = 2(f_7 - y)h_1$$

$$\frac{\partial \ell}{\partial \omega_5} = 2(f_7 - y)h_2$$

$$\frac{\partial \ell}{\partial \beta_2} = 2(f_7 - y)$$

Exercise 6: MLP with Two Inputs

Model:

$$f(x, \theta) = \omega_3 \tanh(\omega_0 x_1 + \omega_1 x_2 + \beta_0) + \beta_1$$
$$\theta = \omega_0, \omega_1, \beta_0, \omega_3, \beta_1$$

6.1 Forward Pass

$$f_0 = \omega_0 x$$

Multiplicacion

$$f_1 = \omega_1 x_2$$

Suma de entradas

$$f_2 = f_0 + f_1$$

Sesgo

$$f_3 = f_2 + \beta_0$$

$$h_1 = \tanh(f_3)$$

Multiplicacion

$$f_4 = \omega_3 h_1$$

Suma final

$$f_5 = f_4 + \beta_1$$

6.2 Loss

$$\ell = (f_5 - y)^2$$

6.3. Forward Computational Graph

$$f_0 = \omega_0 x$$

$$f_1 = \omega_1 x_2$$

$$f_2 = f_0 + f_1$$

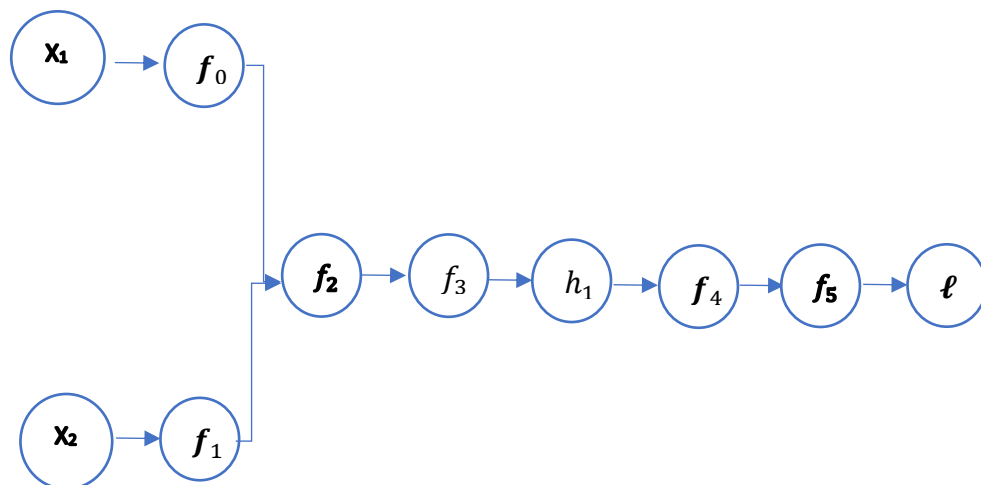
$$f_3 = f_2 + \beta_0$$

$$h_1 = \tanh(f_3)$$

$$f_4 = \omega_3 h_1$$

$$f_5 = f_4 + \beta_1$$

$$\ell = (f_5 - y)^2$$



6.4 Backward Pass

$$\ell = (f_5 - y)^2$$

Derivada

$$\frac{\partial \ell}{\partial f_5} = 2(f_5 - y)$$

Derivada de la suma final

$$f_5 = f_4 + \beta_1$$

$$\frac{\partial f_5}{\partial f_4} = 1$$

$$\frac{\partial f_5}{\partial \beta_1} = 1$$

Regla de la cadena

$$\frac{\partial \ell}{\partial f_4} = 2(f_5 - y)$$

$$\frac{\partial \ell}{\partial \beta_1} = 2(f_5 - y)$$

Multiplicación

Tomando en cuenta que

$$f_4 = \omega_3 h_1$$

Las derivadas son

$$\frac{\partial f_4}{\partial \omega_3} = h_1$$

$$\frac{\partial f_4}{\partial h_1} = \omega_3$$

Regla de la cadena

$$\frac{\partial \ell}{\partial \omega_3} = 2(f_5 - y)h_1$$

$$\frac{\partial \ell}{\partial h_1} = 2(f_5 - y)\omega_3$$

Tanh

$$h_1 = \tanh(f_3)$$

Derivada

$$\frac{\partial h_1}{\partial f_3} = 1 - h_1^2$$

Regla de la cadena

$$\frac{\partial \ell}{\partial f_3} = 2(f_5 - y)\omega_3(1 - h_1^2)$$

Suma

$$f_3 = f_2 + \beta_0$$

$$\frac{\partial f_3}{\partial f_2} = 1$$

$$\frac{\partial f_3}{\partial \beta_0} = 1$$

Regla de la cadena

$$\frac{\partial \ell}{\partial \beta_0} = 2(f_5 - y)\omega_3(1 - h_1^2)$$

$$\frac{\partial \ell}{\partial f_2} = 2(f_5 - y)\omega_3(1 - h_1^2)$$

Multiplicación

$$f_0 = \omega_0 x_1$$

Entonces

$$\frac{\partial f_0}{\partial \omega_0} = x_1$$

$$\frac{\partial f_0}{\partial x_1} = \omega_0$$

Regla de la cadena

$$\frac{\partial \ell}{\partial \omega_0} = 2(f_5 - y)\omega_3(1 - h_1^2)x_1$$

Multiplicación

$$f_1 = \omega_1 x_2$$

Entonces

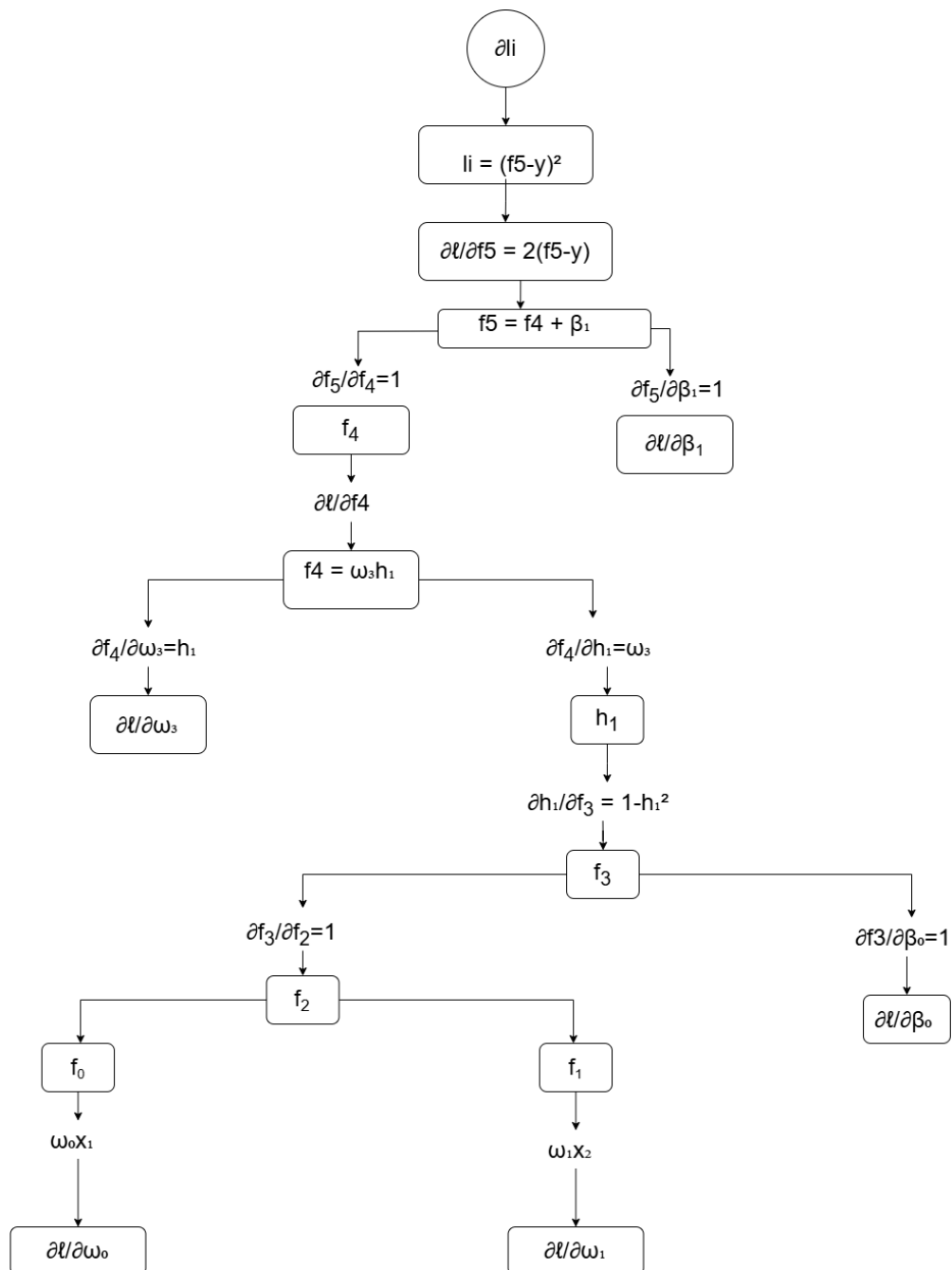
$$\frac{\partial f_0}{\partial \omega_1} = x_2$$

$$\frac{\partial f_0}{\partial x_2} = \omega_1$$

Regla de la cadena

$$\frac{\partial \ell}{\partial \omega_1} = 2(f_5 - y)\omega_3(1 - h_1^2)x_2$$

6.5 Backward Computational Graph



6.6 Compute the gradients with respect to all learnable parameters in θ

$$\theta = \omega_0, \omega_1, \beta_0, \omega_3, \beta_1$$

$$\frac{\partial \ell}{\partial \omega_0} = 2(f_5 - y)\omega_3(1 - h_1^2)x_1$$

$$\frac{\partial \ell}{\partial \omega_1} = 2(f_5 - y)\omega_3(1 - h_1^2)x_2$$

$$\frac{\partial \ell}{\partial \beta_0} = 2(f_5 - y)\omega_3(1 - h_1^2)$$

$$\frac{\partial \ell}{\partial \omega_3} = 2(f_5 - y)h_1$$

$$\frac{\partial \ell}{\partial \beta_1} = 2(f_5 - y)$$

Exercise 7: MLP 2–2–1

Input: $x(x_1, x_2)$

Model:

$$f(x, \theta) = \omega_5 \tanh(\omega_0 x_1 + \omega_1 x_2 + \beta_0) + \omega_6 \text{ReLU}(\omega_2 x_1 + \omega_3 x_2 + \beta_1) + \beta_2$$

$$\theta = \omega_0, \omega_1, \beta_0, \omega_2, \omega_3, \beta_1, \omega_5, \omega_6, \beta_2$$

7.1 Forward Pass

Rama 1

$$f_0 = \omega_0 x_1$$

Multiplicacion

$$f_1 = \omega_1 x_2$$

Suma de entradas

$$f_2 = f_0 + f_1$$

$$f_3 = f_2 + \beta_0$$

$$h_1 = \tanh(f_3)$$

Multiplicacion

$$f_4 = \omega_5 h_1$$

Rama 2 ReLU

$$f_5 = \omega_2 x_1$$

Multiplicacion

$$f_6 = \omega_3 x_2$$

Suma de entradas

$$f_7 = f_5 + f_6$$

$$f_8 = f_7 + \beta_1$$

$$h_2 = \text{ReLU}(f_8)$$

Multiplicacion

$$f_9 = \omega_6 h_2$$

Union

$$f_{10} = f_4 + f_9$$

$$f_{11} = f_{10} + \beta_2$$

7.2 Loss

$$\ell = (f_{11} - y)^2$$

7.3. Forward Computational Graph

$$f_0 = \omega_0 x$$

$$f_1 = \omega_1 x_2$$

$$f_2 = f_0 + f_1$$

$$f_3 = f_2 + \beta_0$$

$$h_1 = \tanh(f_3)$$

$$f_4 = \omega_5 h_1$$

$$f_5 = \omega_2 x_1$$

$$f_6 = \omega_3 x_2$$

$$f_7 = f_5 + f_6$$

$$f_8 = f_7 + \beta_1$$

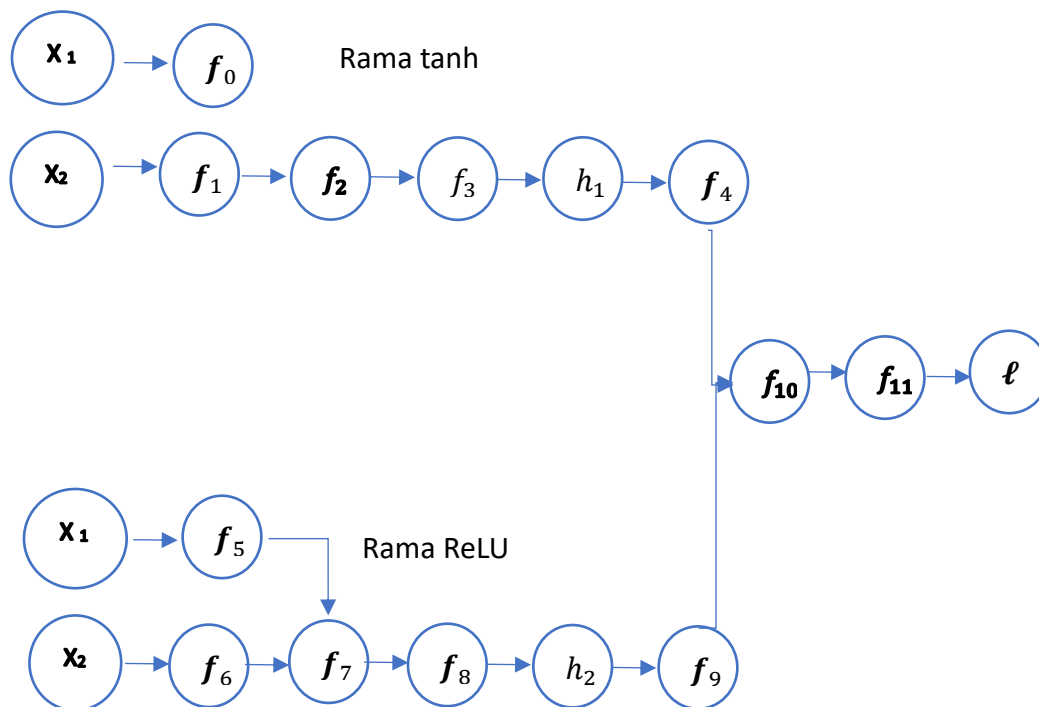
$$h_2 = \text{Relu}(f_8)$$

$$f_9 = \omega_6 h_2$$

$$f_{10} = f_4 + f_9$$

$$f_{11} = f_{10} + \beta_2$$

$$\ell = (f_{11} - y)^2$$



7.4 Backward Pass

$$\ell = (f_{11} - y)^2$$

Derivando

$$\frac{\partial \ell}{\partial f_{11}} = 2(f_{11} - y)$$

Suma

$$f_{11} = f_{10} + \beta_2$$

Regla de la cadena

$$\frac{\partial \ell}{\partial f_{10}} = 2(f_{11} - y)$$

$$\frac{\partial \ell}{\partial \beta_2} = 2(f_{11} - y)$$

Separación de ramas

$$f_{10} = f_4 + f_9$$

Por lo tanto

$$\frac{\partial \ell}{\partial f_4} = 2(f_{11} - y)$$

$$\frac{\partial \ell}{\partial f_9} = 2(f_{11} - y)$$

Rama 1 (tanh)

$$f_4 = \omega_5 h_1$$

$$\frac{\partial f_4}{\partial \omega_5} = h_1$$

$$\frac{\partial f_4}{\partial h_1} = \omega_5$$

Regla de la cadena

$$\frac{\partial \ell}{\partial \omega_5} = 2(f_{11} - y)h_1$$

$$\frac{\partial \ell}{\partial h_1} = 2(f_{11} - y)\omega_5$$

Tanh

$$h_1 = \tanh(f_3)$$

Derivada

$$\frac{\partial h_1}{\partial f_3} = 1 - h_1^2$$

Regla de la cadena

$$\frac{\partial \ell}{\partial f_3} = 2(f_{11} - y)\omega_5(1 - h_1^2)$$

Suma

$$f_3 = f_2 + \beta_0$$

$$\frac{\partial \ell}{\partial \beta_0} = 2(f_{11} - y)\omega_5(1 - h_1^2)$$

$$\frac{\partial \ell}{\partial f_2} = 2(f_{11} - y)\omega_5(1 - h_1^2)$$

Suma de entradas

$$f_2 = f_0 + f_1$$

$$\frac{\partial \ell}{\partial f_0} = 2(f_{11} - y)\omega_5(1 - h_1^2)$$

$$\frac{\partial \ell}{\partial f_1} = 2(f_{11} - y)\omega_5(1 - h_1^2)$$

Multiplicacion

$$\frac{\partial \ell}{\partial \omega_0} = 2(f_{11} - y)\omega_5(1 - h_1^2)x_1$$

$$\frac{\partial \ell}{\partial \omega_1} = 2(f_{11} - y)\omega_5(1 - h_1^2)x_2$$

Rama 2 (ReLU)

$$f_9 = \omega_6 h_2$$

$$\frac{\partial f_9}{\partial \omega_6} = h_2$$

$$\frac{\partial f_9}{\partial h_2} = \omega_6$$

Regla de la cadena

$$\frac{\partial \ell}{\partial \omega_6} = 2(f_{11} - y)h_2$$

$$\frac{\partial \ell}{\partial h_2} = 2(f_{11} - y)\omega_6$$

ReLU

$$h_2 = \text{ReLU}(f_8)$$

$$\frac{\partial h_2}{\partial f_8} = \begin{cases} -1, & f_8 < 0 \\ 0, & f_8 \geq 0 \end{cases}$$

Regla de la cadena

$$\frac{\partial \ell}{\partial f_8} = \frac{\partial \ell}{\partial h_2} \cdot \frac{\partial h_2}{\partial f_8}$$

Suma

$$f_8 = f_7 + \beta_1$$

$$\frac{\partial \ell}{\partial \beta_1} = \frac{\partial \ell}{\partial f_8}$$

$$\frac{\partial \ell}{\partial f_7} = \frac{\partial \ell}{\partial f_8}$$

Suma de entradas

$$f_7 = f_5 + f_6$$

$$\frac{\partial \ell}{\partial f_5} = \frac{\partial \ell}{\partial f_8}$$

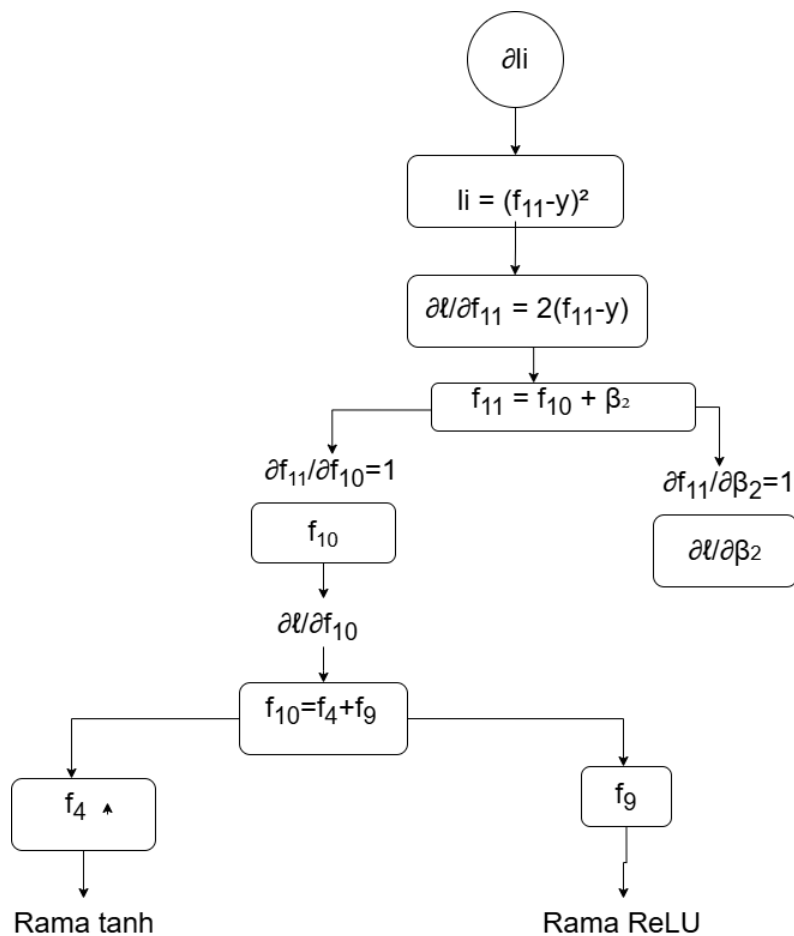
$$\frac{\partial \ell}{\partial f_6} = \frac{\partial \ell}{\partial f_8}$$

Multiplicaciones

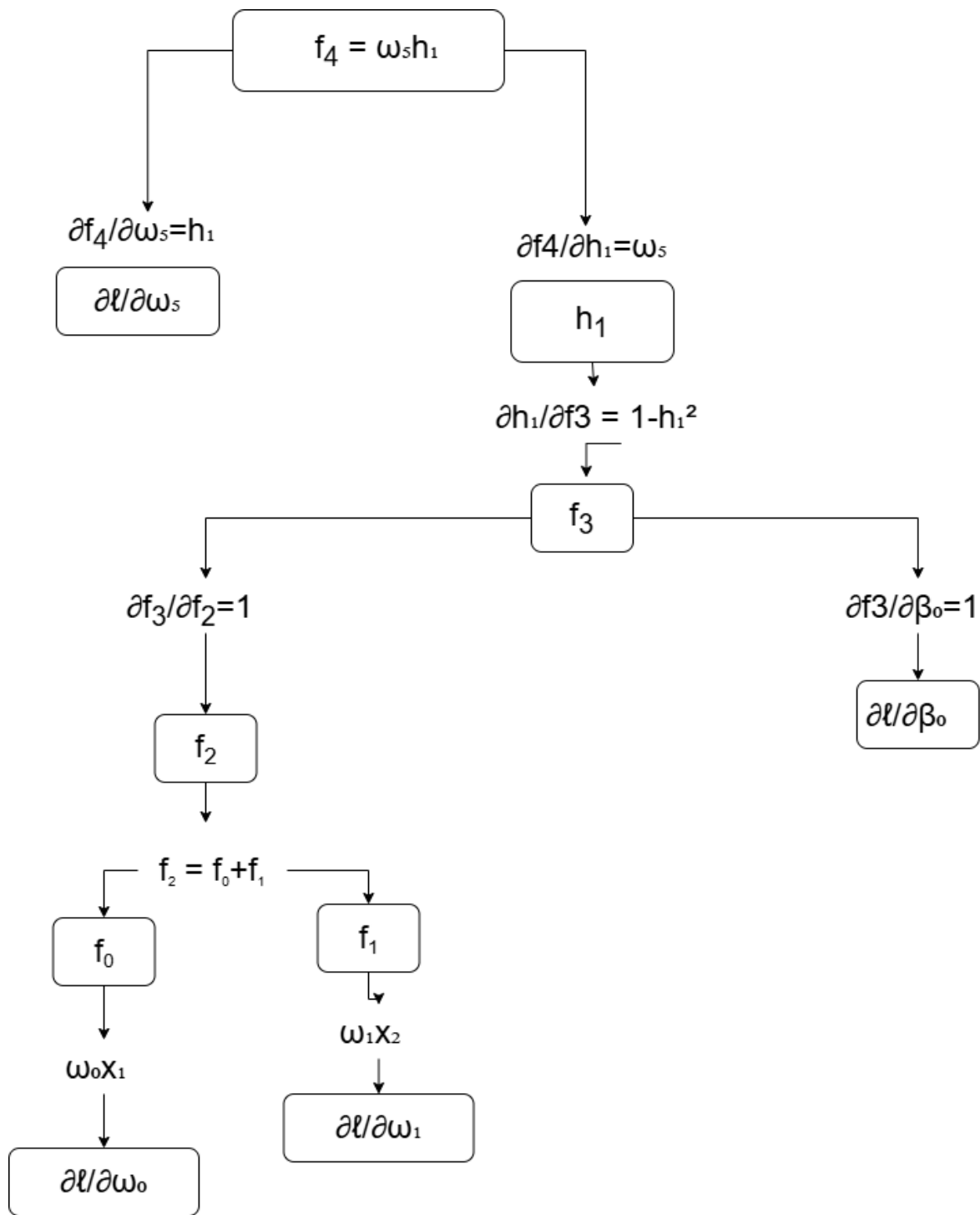
$$\frac{\partial \ell}{\partial \omega_2} = \frac{\partial \ell}{\partial f_8} x_1$$

$$\frac{\partial \ell}{\partial \omega_3} = \frac{\partial \ell}{\partial f_8} x_2$$

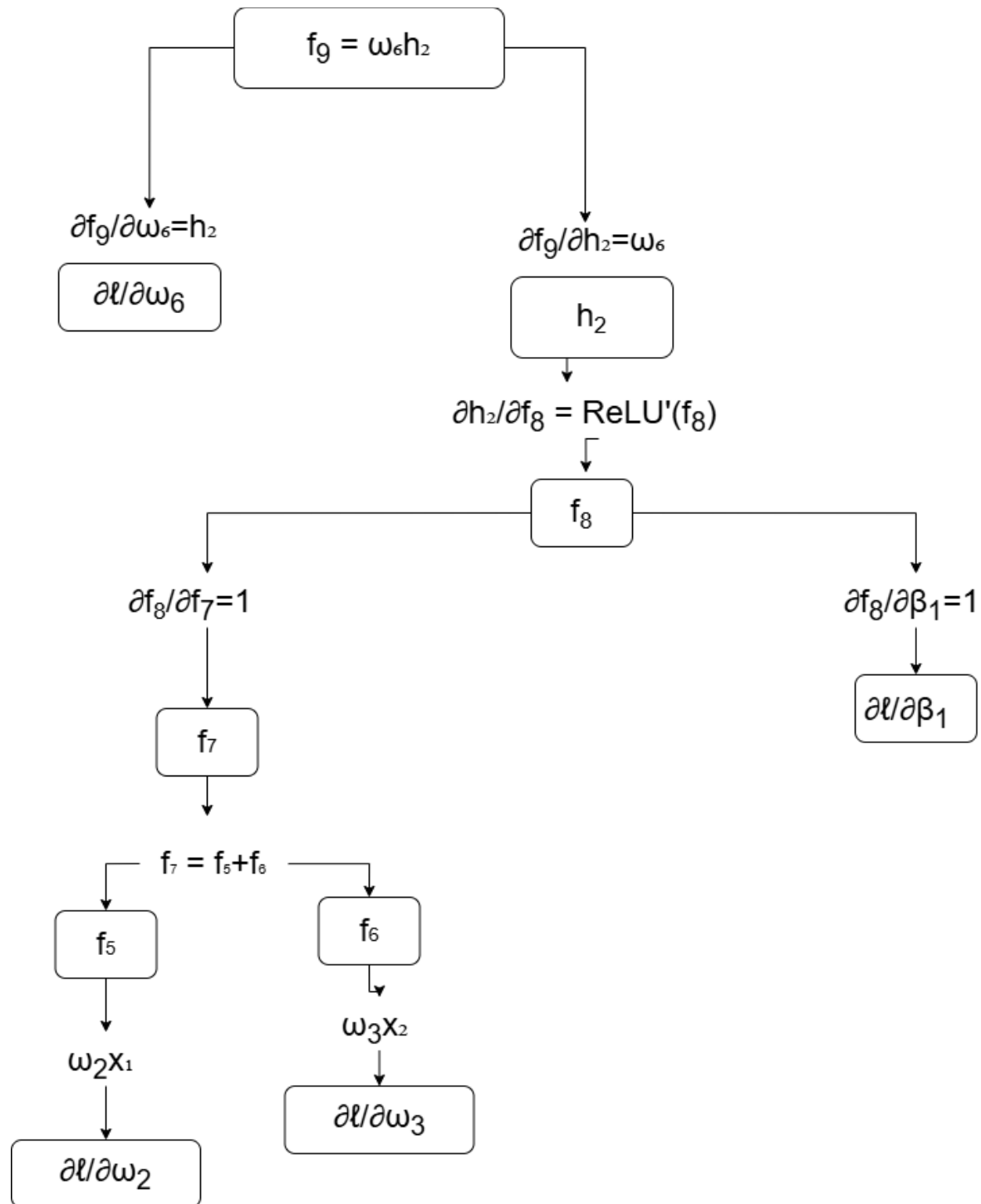
7.5 Backward Computational Graph



Rama 1 (tanh)



Rama 2 (ReLU)



7.6 Compute the gradients with respect to all learnable parameters in θ

$$\theta = \omega_0, \omega_1, \beta_0, \omega_2\omega_3, \beta_1, \omega_5, \omega_6, \beta_2$$

$$\frac{\partial \ell}{\partial \omega_0} = 2(f_{11} - y)\omega_5(1 - h_1^2)x_1$$

$$\frac{\partial \ell}{\partial \omega_1} = 2(f_{11} - y)\omega_5(1 - h_1^2)x_2$$

$$\frac{\partial \ell}{\partial \beta_0} = 2(f_{11} - y)\omega_5(1 - h_1^2)$$

$$\frac{\partial \ell}{\partial \omega_2} = \frac{\partial \ell}{\partial f_8} x_1$$

$$\frac{\partial \ell}{\partial \omega_3} = \frac{\partial \ell}{\partial f_8} x_2$$

$$\frac{\partial \ell}{\partial \beta_1} = \frac{\partial \ell}{\partial f_8}$$

$$\frac{\partial \ell}{\partial \omega_5} = 2(f_{11} - y)h_1$$

$$\frac{\partial \ell}{\partial \omega_6} = 2(f_{11} - y)h_2$$

$$\frac{\partial \ell}{\partial \beta_2} = 2(f_{11} - y)$$

Exercise 8: Branching Model with Shared Pre-Activation

Model:

$$f(x, \theta) = \omega_3 \sin(\omega_0 x + \beta_0) + \omega_4 \cos(\omega_0 x + \beta_0) + \beta_1$$

$$\theta = \omega_0, \beta_0, \omega_3, \omega_4, \beta_1$$

8.1 Forward Pass

Multiplicación

$$f_1 = \omega_1 x_2$$

Suma (nodo compartido)

$$f_1 = f_0 + \beta_0$$

Rama Seno

$$h_1 = \sin(f_1)$$

$$f_2 = \omega_3 h_1$$

Rama coseno

$$h_2 = \cos(f_1)$$

$$f_3 = \omega_4 h_2$$

Union

$$f_4 = f_2 + f_3$$

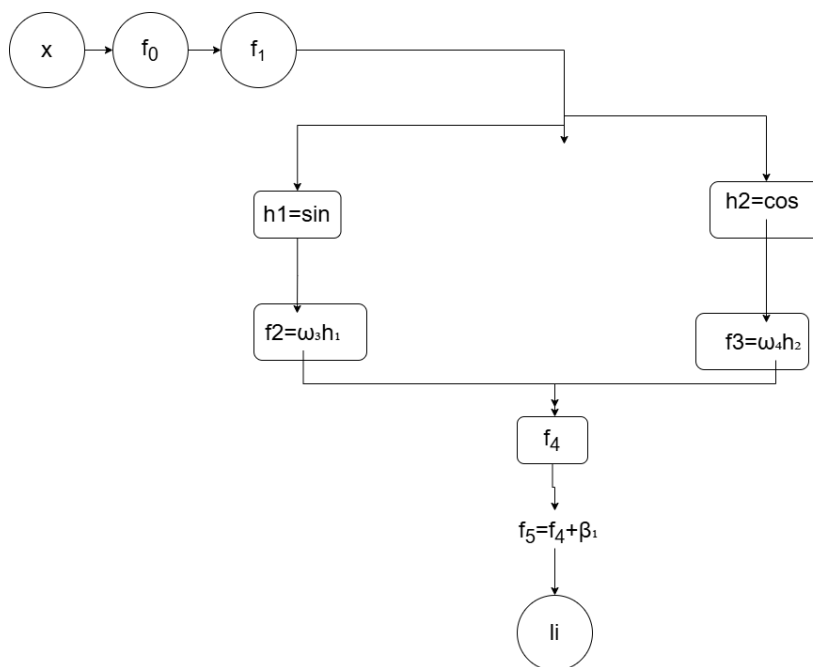
$$f_5 = f_4 + \beta_1$$

8.2 Loss

$$\ell = (f_5 - y)^2$$

8.3. Forward Computational Graph

$$\begin{aligned} f_0 &= \omega_0 x \\ f_1 &= f_0 + \beta_0 \\ h_1 &= \sin(f_1) \\ f_2 &= \omega_3 h_1 \\ h_2 &= \cos(f_1) \\ f_3 &= \omega_4 h_2 \\ f_4 &= f_2 + f_3 \\ f_5 &= f_4 + \beta_1 \\ \ell &= (f_5 - y)^2 \end{aligned}$$



8.4 Backward Pass

$$\ell = (f_5 - y)^2$$

$$\frac{\partial \ell}{\partial f_5} = 2(f_5 - y)$$

Derivada de la suma final

$$f_5 = f_4 + \beta_1$$

$$\frac{\partial \ell}{\partial f_4} = 2(f_5 - y)$$

$$\frac{\partial \ell}{\partial \beta_1} = 2(f_5 - y)$$

Separacion de ramas

$$f_4 = f_2 + f_3$$

$$\frac{\partial \ell}{\partial f_2} = 2(f_5 - y)$$

$$\frac{\partial \ell}{\partial f_3} = 2(f_5 - y)$$

Rama Seno

$$f_2 = \omega_3 h_1$$

$$\frac{\partial f_2}{\partial \omega_3} = h_1$$

$$\frac{\partial f_2}{\partial h_1} = \omega_3$$

Regla de la cadena

$$\frac{\partial \ell}{\partial \omega_3} = 2(f_5 - y)h_1$$

$$\frac{\partial \ell}{\partial h_1} = 2(f_5 - y)\omega_3$$

Seno

$$h_1 = \sin(f_1)$$

$$\frac{\partial h_1}{\partial f_1} = \cos(f_1)$$

Regla de la cadena

$$\frac{\partial \ell}{\partial f_1} \sin = 2(f_5 - y)\omega_3 \cos(f_1)$$

Rama coseno

$$f_3 = \omega_4 h_2$$

$$\frac{\partial f_3}{\partial \omega_4} = h_2$$

$$\frac{\partial f_3}{\partial h_2} = \omega_4$$

$$\frac{\partial \ell}{\partial \omega_4} = 2(f_5 - y)h_2$$

$$\frac{\partial \ell}{\partial h_2} = 2(f_5 - y)\omega_4$$

Coseno

$$h_2 = \cos(f_1)$$

$$\frac{\partial h_2}{\partial f_1} = -\sin(f_1)$$

Regla de la cadena

$$\frac{\partial \ell}{\partial f_1} \cos = -2(f_5 - y)\omega_4 \cos(f_1)$$

Acumulación de Gradientes

Dado

$$f_1 = \omega_0 x + \beta_0$$

$$\frac{\partial \ell}{\partial f_1} = \frac{\partial \ell}{\partial f_1} \sin + \frac{\partial \ell}{\partial f_1} \cos$$

$$\frac{\partial \ell}{\partial f_1} = 2(f_5 - y)[\omega_3 \cos(f_1) - \omega_4 \sin(f_1)]$$

Nodo compartido

$$f_1 = f_0 + \beta_0$$

$$\frac{\partial \ell}{\partial \beta_0} = \frac{\partial \ell}{\partial f_1}$$

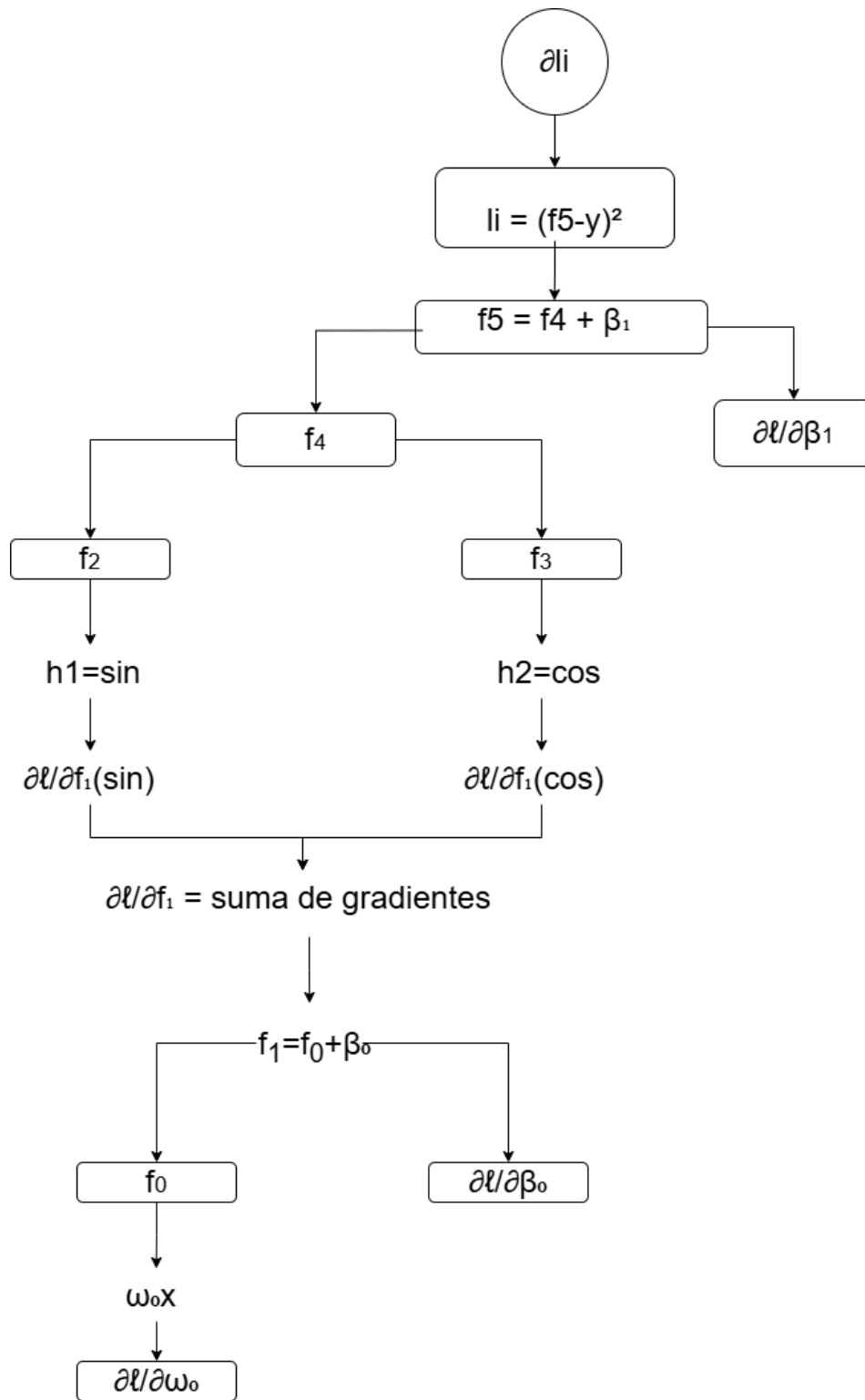
$$\frac{\partial \ell}{\partial f_0} = \frac{\partial \ell}{\partial f_1}$$

Multiplicación inicial

$$f_0 = \omega_0 x$$

$$\frac{\partial \ell}{\partial \omega_0} = \frac{\partial \ell}{\partial f_1} x$$

8.5 Backward Computational Graph



8.6 Compute the gradients with respect to all learnable parameters in θ

$$\theta = \omega_0, \beta_0, \omega_3, \omega_4, \beta_1,$$

$$\frac{\partial \ell}{\partial \omega_0} = 2(f_5 - y)[\omega_3 \cos(f_1) - \omega_4 \sin(f_1)]x$$

$$\frac{\partial \ell}{\partial \beta_0} = 2(f_5 - y)[\omega_3 \cos(f_1) - \omega_4 \sin(f_1)]$$

$$\frac{\partial \ell}{\partial \omega_3} = 2(f_5 - y)h_1$$

$$\frac{\partial \ell}{\partial \omega_4} = 2(f_5 - y)h_2$$

$$\frac{\partial \ell}{\partial \beta_1} = 2(f_5 - y)$$